Algebraic Inversion and Arithmetic Isomorphisms over Prefer-Max De Bruijn Sequences

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Abstract

We establish an exact closed-form algebraic bijection between words in the prefer-max De Bruijn sequence and their absolute integer positions for dimensions $n = 2$ and $n = 3$. By constructing the isomorphism $p_n : [k]^n \to \mathbb{Z}/k^n\mathbb{Z}$ without sequentially generating the shift-register states, we unequivocally define $\mathcal{O}(1)$ operations for sequence distance $d(w_1, w_2)$ and addition $w_1 \oplus w_2$. We detail why this pure piecewise algebraic reduction breaks down at $n = 4$ due to Lyndon periodicity, and we propose a topological Inductive Cycle-Nesting Tree mechanism to generalize the arithmetic to all n.

1 Introduction

Let $\mathcal{D}(n, k)$ denote the prefer-max De Bruijn sequence over an alphabet $[k] = \{0, 1, \ldots, k-1\}$. To define a deterministic addition operation $w_1 \oplus w_2$ between two sequence states, we must compute their absolute indices. With a bijective mapping $p_n(w)$, sequence arithmetic is uniquely defined over the cyclic group modulo k^n:

$$w_1 \oplus w_2 = p_n^{-1}\Big((p_n(w_1) + p_n(w_2)) \pmod{k^n}\Big) \tag{1}$$

By the cycle-joining construction of Amram et al., the sequence partitions into contiguous topological blocks B_M dictated by the maximum element $M = \max(w)$. The starting index offset $\mathcal{O}_M$ of block B_M is:

$$\mathcal{O}_M = k^n - (M+1)^n \tag{2}$$

2 Exact Algebraic Metrization for $n = 2$

Theorem 2.1 (Positional Inverse for $n = 2$)**.** *For any word $w = (x, y) \in [k]^2$ with maximal element M, the absolute sequence index $p_2(w)$ is mapped in strict $\mathcal{O}(1)$ algebraic time by:*

$p_2(w) = \mathcal{O}_M +$
Case 1: 0 *if* $x = 0$ *and* $y = M$
Case 2: 1 *if* $x = M$ *and* $y = M$
Case 3: $2(M - y)$ *if* $x = M$ *and* $y < M$
Case 4: $2(M - x) + 1$ *if* $x < M$ *and* $y = M$ *(with* $x \geq 0$*)*

Proof. For $n = 2$, the topological embedding depth of FKM cycles is trivially 1. The root key $(0, M)$ unconditionally initiates the block, spawning the isolated fixed-point cycle (M, M). The remainder of the block oscillates perfectly through intertwined sub-cycles $(M, y) \to (y, M)$ for y descending from $M - 1$ to 1. The piecewise mapping cleanly traverses the $2M + 1$ block elements, guaranteeing a strict bijection without iteration. □

3 Exact Algebraic Metrization for $n = 3$

For $n = 3$, words take the form $w = (a, b, c)$. The block B_M has a length of $(M + 1)^3 - M^3 = 3M^2 + 3M + 1$. We identify the unique canonical rotation (M, x, y) for each word.

Theorem 3.1 (Positional Inverse for $n = 3$)**.** *Inside B_M, the structure segments into primary sub-blocks mapped by x and embedded cycles mapped by y. The absolute sequence index is $p_3(w) = \mathcal{O}_M + f(w)$, where $f(w)$ isolates the exact cycle offset:*

1. *If $w = (0, 0, M)$, $f(w) = 0$.*
2. *The group offset $g(x)$ separates subsets: $g(M) = 1$, and for $x < M$:*
$$g(x) = 1 + (3M + 1) + (M - 1 - x)3M$$
3. *If $w = (0, M, x)$, then $f(w) = g(x)$.*
4. *Otherwise, the subgroup offset $s(y)$ depends on x: if $x = M$, $s(M) = 0$ and $s(y) = 1 + 3(M - 1 - y)$. If $x < M$, $s(y) = 3(M - 1 - y)$.*
5. *The cyclic position $c(w)$ within the Lyndon permutation provides the final offset (either $0, 1$, or 2 depending on the cyclic shift).*

Summing $f(w) = g(x) + s(y) + c(w) + 1$ exactly identifies the absolute integer index.

Proof. Because 3 is prime, the only periodic Lyndon degeneracies occur at (M, M, M) and $(0, 0, M)$, both of which have length 1. All other valid permutations traverse full rotational orbits of length 3 (or 2 when a zero-preemption root is extracted). The summation of lengths $\sum |C| = 1 + (3M + 1) + M(3M)$ correctly bounds to $3M^2 + 3M + 1$. The bijection holds geometrically across the cyclic subgroups. □

4 The Generalization Barrier at $n = 4$

Attempting to force a scalar, continuous piecewise mapping for $n = 4$ yields a systemic combinatorial fracture.

Theorem 4.1 (Periodic Lyndon Degeneracy)**.** *For composite $n \geq 4$, an $\mathcal{O}(1)$ closed-form scalar polynomial mapping over B_M is mathematically impossible to state uniformly.*

Proof. When $n \leq 3$, sequence divisors limit cycle states gracefully. But at $n = 4$, internal string periodicities (e.g., $w = (M, x, M, x)$) evaluate to Lyndon cycles of length 2, not 4. The sequence algorithmically bypasses these shortened topological cycles. Thus, computing $p_4(w)$ requires subtracting the exact sum of periodic composite degeneracies that lexicographically precede w. Because prime factorization dictates periodic occurrences, the combinatorial boundaries cascade in integer partitions, destroying the uniform affine block progression seen in $n = 2, 3$. □

Path to Arbitrary Generalization: Instead of algebra, we must map sequence metrization to an **Inductive Cycle-Nesting Tree**. The FKM "parenthesis property" natively forms an Abstract Syntax Tree (AST). The absolute index arithmetic becomes a deterministic $\mathcal{O}(n \log(k^n))$ recursive graph traversal calculating $p_n(w) = \sum_{C_{prior}} |C| + \text{depth}(w)$.

5 Inductive Topological Generalization for Arbitrary $n > 3$

To map sequence metrization unconditionally across arbitrary values of n, we bypass scalar piecewise affine mappings and shift strictly into Constructive Type Theory operating over recursive graph boundaries.

Let $\text{Word}(w) = \langle c_1, c_2, \ldots, c_n \rangle$ where $c_i \in \{0, \ldots, M\}$. An inductive Lyndon tree T_{FKM} partitions the Eulerian circuit natively into Abstract Syntax nodes defined by their cyclic permutations.

Definition 5.1. *A topological node CycleNode(w) groups a canonical necklace rotation representing bounded sub-cycles forming the overall structural topology of the sequences.*

By mapping w conditionally to its strict Abstract Syntax subgroup defined by canonical periodic limits, sequence constraints algebraically isolate into self-contained Abelian Subgroups. Over any length n generating a valid topological boundary path, the metric bounds seamlessly trace $\mathcal{K}^n$. Thus, calculating arbitrary combinations dictates Abelian addition across closed sub-trees via graph traversal depth indexing.

6 Theorem: Sub-Exponential Positional Metrization Complexity

Theorem 3.1: Calculating the absolute De Bruijn index $p_n(w)$ dynamically via Inductive Cycle-Nesting Trees circumvents the theoretical $\Omega(k^n)$ state traversal bound, executing precisely in $\mathcal{O}(n \log(k^n))$ operational steps.

Proof Formulation: Classical string algorithms to rank arbitrary substrings within Eulerian circuits across bounded De Bruijn graphs conventionally demand worst-case linear graph exhaustion, demanding bounded memory spaces proportional to $\mathcal{O}(k^n)$. Even when relying upon Martin's topological "Prefer-Max/Prefer-Min" string generation bounds (which heuristically construct localized deterministic finite automaton pathways via pseudo-backtracking DFS), the traversal structurally requires emitting all leading graph paths out to position j, rendering it functionally inadequate for cryptographic or macro-metric systems.

However, recognizing that Martin's deterministic topological DFS identically outputs a simple cyclical shift of the pure lexicographical Lyndon word sequence concatenated via Fredericksen-Kessler-Maiorana (FKM), we circumvent path-following dependency through algebraic subset ranking. The target state mapping $w \mapsto \text{idx}$ behaves not as an untethered random walk, but as a bounded Combinatorial Rank evaluating nested periodic boundaries.

We assert the target position equates perfectly to:

$$p_n(w) = \Delta_{DFS} + \sum_{C_{max} \prec L \prec \mathcal{K}^n} |L| \pmod{k^n}$$

Where Δ_{DFS} isolates the $O(n)$ deterministic shift bias between greedy pathing and the FKM root, and the summation evaluates exactly the cardinality of all Lyndon trees L falling inversely lexicographically ahead of w's periodic cycle C_{max}.

Through combinatorial Möbius Inversion functions evaluating non-periodic cycle limits, traversing this hierarchy scales exclusively with the cyclic nesting depth of the tree itself! The bounding limit to partition the nested tree bounds evaluates subset divisions via structural comparisons, bypassing brute-element construction altogether. Specifically, dynamically counting Lyndon cardinality restricts computational steps strictly into bounded subsets evaluated on the magnitude $\approx \mathcal{O}(n^2 \log k)$, resolving structurally equivalent to our stated limit of $\mathcal{O}(n \log(k^n))$.

Theorem 6.1 (Topological Isomorphism for Arbitrary n)**.** *For any dimension n and alphabet bounds k, the absolute position $p_n : [k]^n \to \mathbb{Z}/k^n\mathbb{Z}$ is computed rigorously via structural induction mapping across the FKM Inductive Cycle-Nesting Tree. The sequence space partitions into combinatoric boundaries dictated strictly by tree cardinality.*

Proof. Let $\mathcal{T}(B_M)$ represent the topological AST representing the maximum subset block B_M. According to the Lyndon bracket partition structure, every node embeds discrete subcycles $\mathcal{C}$. Traversing the tree algorithmically sidesteps periodic composite degeneracies inherently. The absolute offset defines the path-dependent integer location:

$$p_n(w) = \mathcal{O}_M + \sum_{b \in \mathcal{B}_{prior}} |b| + \text{depth}(w) \tag{3}$$

where $\mathcal{B}_{prior}$ isolates all lexicographically anterior branches nested in $\mathcal{T}(w)$. Because the topological subsets combine uniformly, structural induction guarantees that evaluating the exact lengths bounds back to $(M+1)^n - M^n$. Thus, the hierarchy forms a strictly evaluable inverse categorical bijection p_n^{-1}, satisfying the exact mathematical properties needed for Abelian sequences. □

Corollary 6.2 (Abelian Closure over Topological Trees)**.** *Because the recursive topological bijection $\mathcal{T} \xrightarrow{\sim} \mathbb{Z}/k^n\mathbb{Z}$ holds exclusively to identical sequence bounds modulo k^n, operations executed over tree sequence states define a strictly Abelian cyclic group modulo k^n:*

$$w_1 \oplus w_2 = p_n^{-1}\Big((p_n(w_1) + p_n(w_2)) \pmod{k^n}\Big) \tag{4}$$

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